

Презентация по лабораторной работе 4

Первоначальное конфигурирование сети

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1 Информация

1.1 Докладчик

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2 Вводная часть

2.1 Цели и задачи

- Провести подготовительную работу по первоначальной настройке коммутаторов сети.

3 Выполнение заданий

3.1 слайд 1

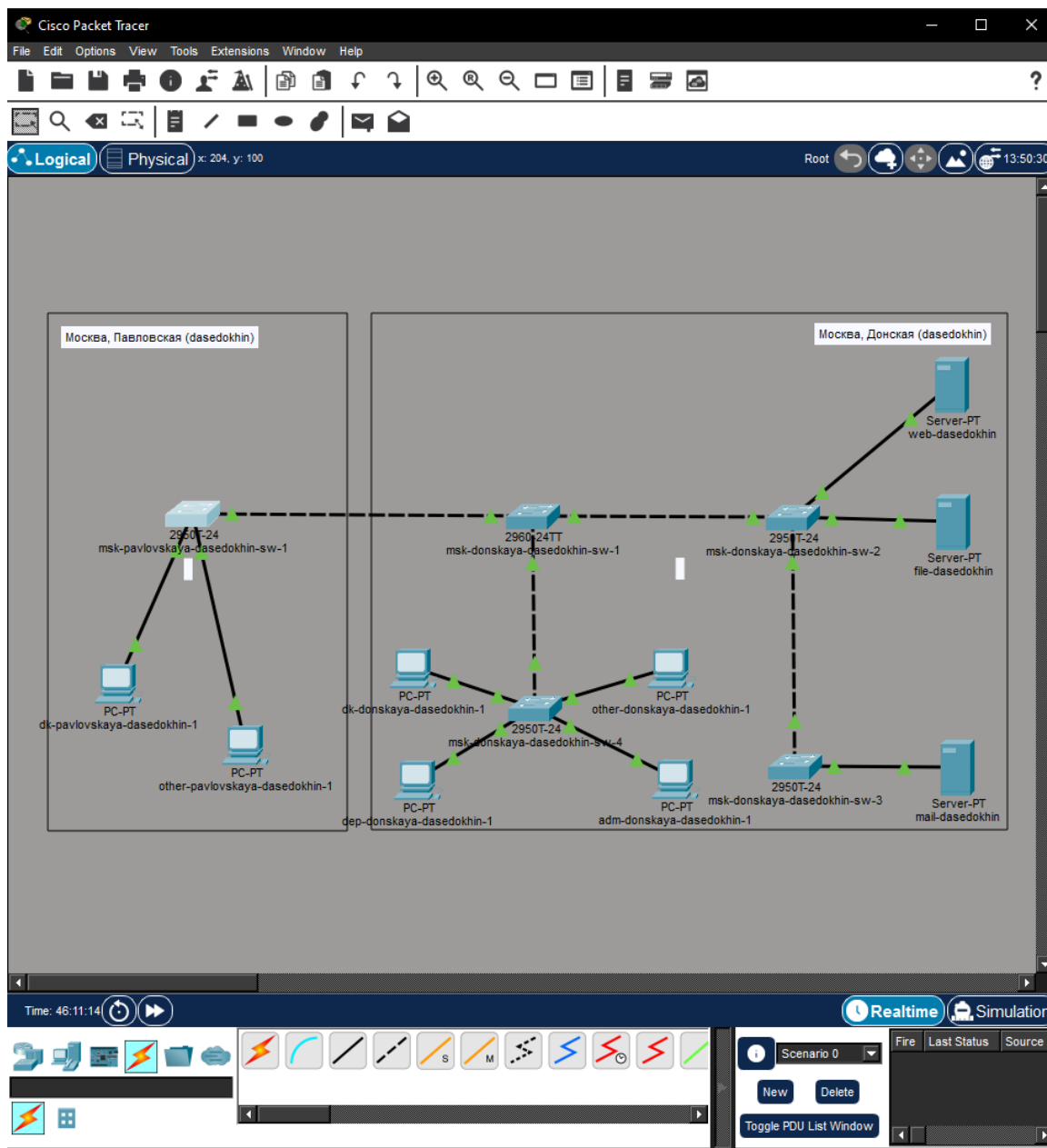


Рисунок 3.1: Размещение коммутаторов и оконечных устройств согласно схеме

3.2 слайд 2

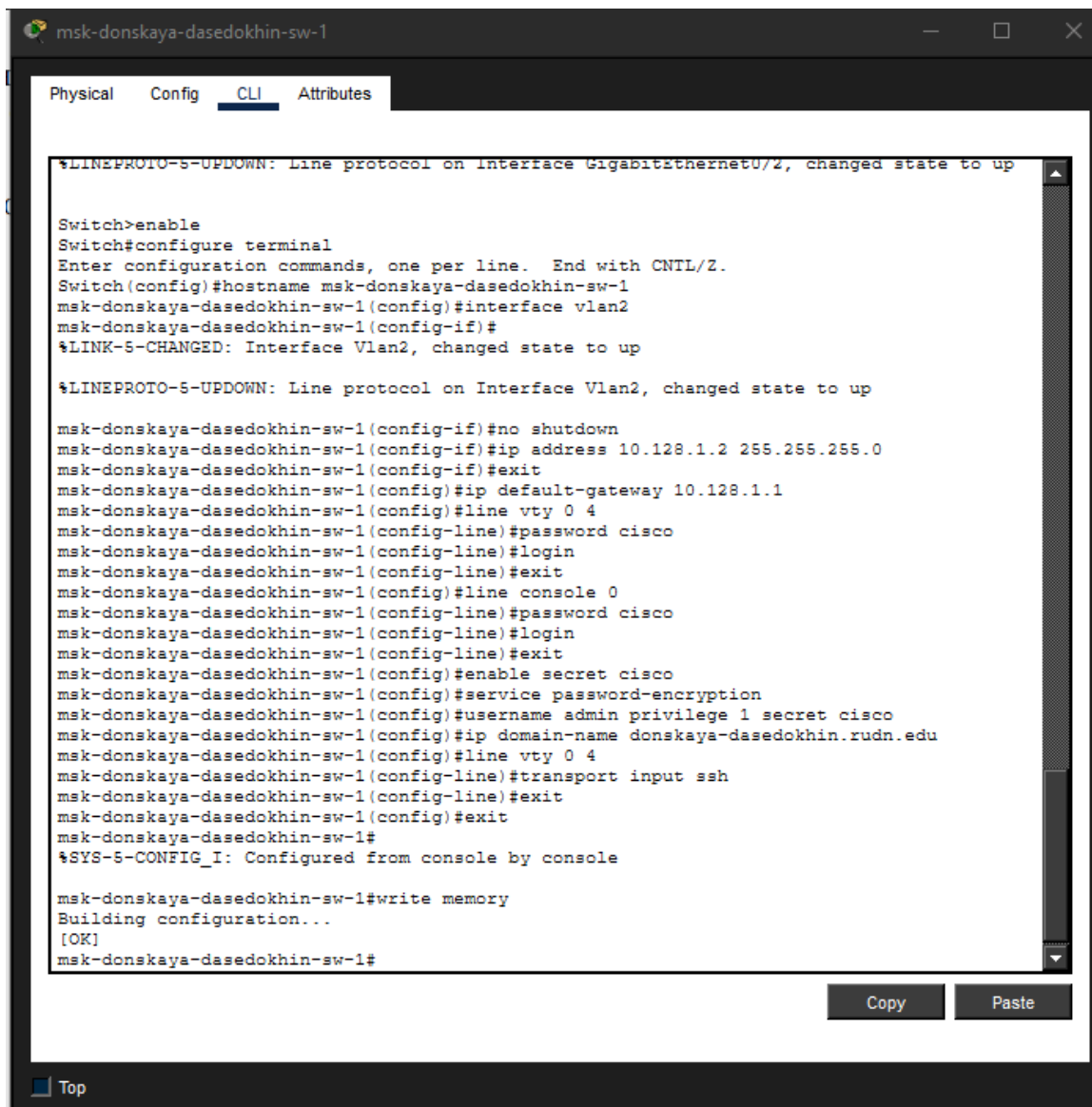


Рисунок 3.2: Настройка коммутатора msk-donskaya-dasedokhin-sw-1

3.3 слайд 3



```
msk-donskaya-dasedokhin-sw-2
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dasedokhin-sw-2
msk-donskaya-dasedokhin-sw-2(config)#interface vlan2
msk-donskaya-dasedokhin-sw-2(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dasedokhin-sw-2(config-if)#no shutdown
msk-donskaya-dasedokhin-sw-2(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dasedokhin-sw-2(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dasedokhin-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-dasedokhin-sw-2(config-if)#exit
msk-donskaya-dasedokhin-sw-2(config)#ip default gateway 10.128.1.1
msk-donskaya-dasedokhin-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-dasedokhin-sw-2(config)#line vty 0 4
msk-donskaya-dasedokhin-sw-2(config-line)#password cisco
msk-donskaya-dasedokhin-sw-2(config-line)#login
msk-donskaya-dasedokhin-sw-2(config-line)#exit
msk-donskaya-dasedokhin-sw-2(config)#line console 0
msk-donskaya-dasedokhin-sw-2(config-line)#password cisco
msk-donskaya-dasedokhin-sw-2(config-line)#login
msk-donskaya-dasedokhin-sw-2(config-line)#exit
msk-donskaya-dasedokhin-sw-2(config)#enable secret cisco
msk-donskaya-dasedokhin-sw-2(config)#service password-encryption
msk-donskaya-dasedokhin-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-dasedokhin-sw-2(config)#ip domain name donskeya-dasedokhin.rudn.edu
msk-donskaya-dasedokhin-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dasedokhin-sw-2.donskeya-dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dasedokhin-sw-2(config)#line vty 0 4
*Mar 2 22:20:43.704: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 2 22:20:43.704: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dasedokhin-sw-2(config-line)#transport input ssh
msk-donskaya-dasedokhin-sw-2(config-line)#exit
msk-donskaya-dasedokhin-sw-2(config)#exit
msk-donskaya-dasedokhin-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

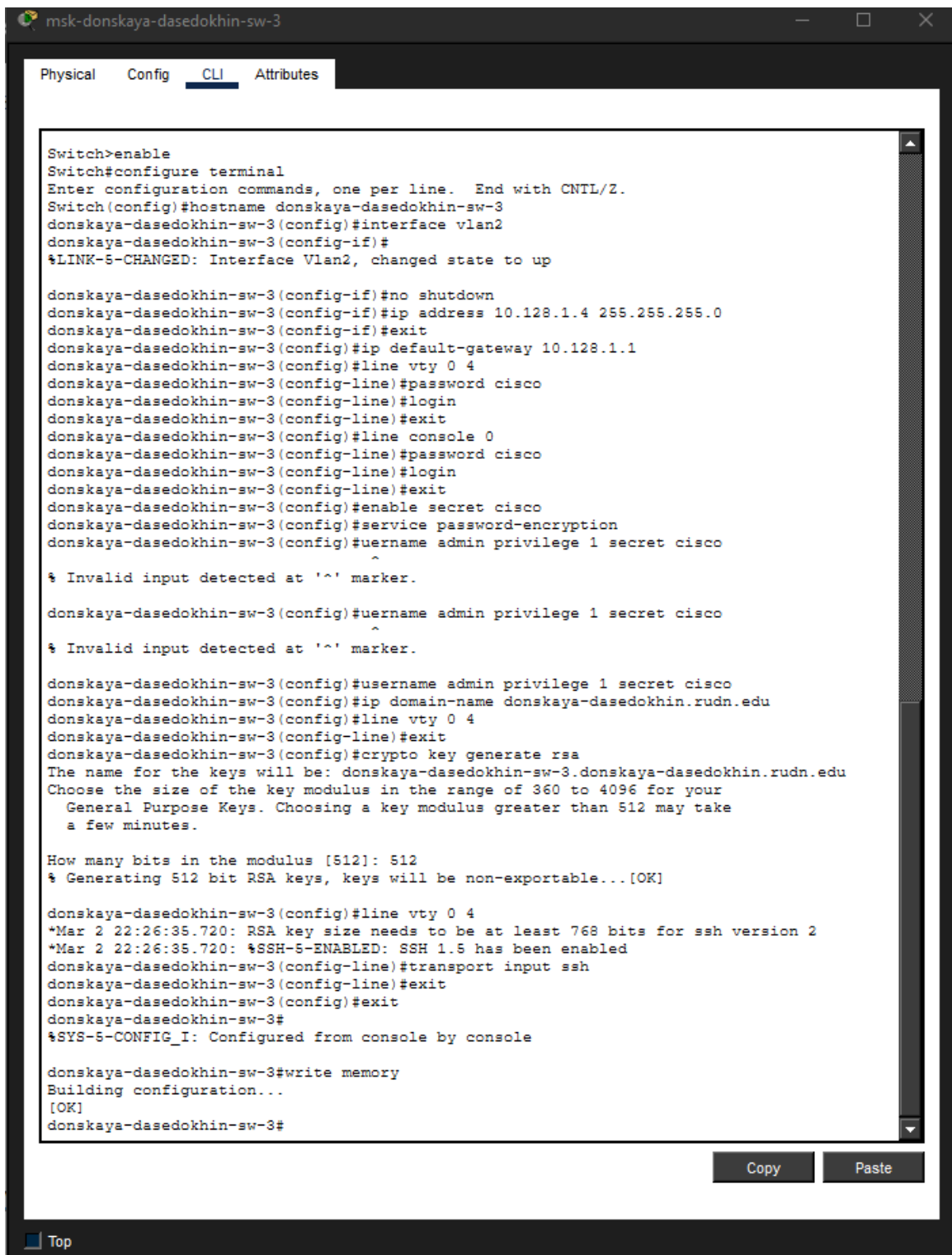
msk-donskaya-dasedokhin-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-dasedokhin-sw-2#
```

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Рисунок 3.3: Настройка коммутатора msk-donskaya-dasedokhin-sw-2

3.4 слайд 4



```
msk-donskaya-dasedokhin-sw-3
Physical Config CLI Attributes

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname donsokaya-dasedokhin-sw-3
donsokaya-dasedokhin-sw-3(config)#interface vlan2
donsokaya-dasedokhin-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

donsokaya-dasedokhin-sw-3(config-if)#no shutdown
donsokaya-dasedokhin-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
donsokaya-dasedokhin-sw-3(config-if)#exit
donsokaya-dasedokhin-sw-3(config)#ip default-gateway 10.128.1.1
donsokaya-dasedokhin-sw-3(config)#line vty 0 4
donsokaya-dasedokhin-sw-3(config-line)#password cisco
donsokaya-dasedokhin-sw-3(config-line)#login
donsokaya-dasedokhin-sw-3(config-line)#exit
donsokaya-dasedokhin-sw-3(config)#line console 0
donsokaya-dasedokhin-sw-3(config-line)#password cisco
donsokaya-dasedokhin-sw-3(config-line)#login
donsokaya-dasedokhin-sw-3(config-line)#exit
donsokaya-dasedokhin-sw-3(config)#enable secret cisco
donsokaya-dasedokhin-sw-3(config)#service password-encryption
donsokaya-dasedokhin-sw-3(config)#username admin privilege 1 secret cisco
^
% Invalid input detected at '^' marker.

donsokaya-dasedokhin-sw-3(config)#username admin privilege 1 secret cisco
^
% Invalid input detected at '^' marker.

donsokaya-dasedokhin-sw-3(config)#username admin privilege 1 secret cisco
donsokaya-dasedokhin-sw-3(config)#ip domain-name donsokaya-dasedokhin.rudn.edu
donsokaya-dasedokhin-sw-3(config)#line vty 0 4
donsokaya-dasedokhin-sw-3(config-line)#exit
donsokaya-dasedokhin-sw-3(config)#crypto key generate rsa
The name for the keys will be: donsokaya-dasedokhin-sw-3.donsokaya-dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

donsokaya-dasedokhin-sw-3(config)#line vty 0 4
*Mar 2 22:26:35.720: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 2 22:26:35.720: %SSH-5-ENABLED: SSH 1.5 has been enabled
donsokaya-dasedokhin-sw-3(config-line)#transport input ssh
donsokaya-dasedokhin-sw-3(config-line)#exit
donsokaya-dasedokhin-sw-3(config)#exit
donsokaya-dasedokhin-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

donsokaya-dasedokhin-sw-3#write memory
Building configuration...
[OK]
donsokaya-dasedokhin-sw-3#
```

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Рисунок 3.4: Настройка коммутатора msk-donskaya-dasedokhin-sw-3

3.5 слайд 5



The screenshot shows a network switch configuration window titled "msk-donskaya-dasedokhin-sw-4". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The CLI window displays the following commands and their outputs:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#configure terminal
^
% Invalid input detected at '^' marker.

Switch(config)#hostname msk-donskaya-dasedokhin-sw-4
msk-donskaya-dasedokhin-sw-4(config)#interface vlan2
msk-donskaya-dasedokhin-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dasedokhin-sw-4(config-if)#no shutdown
msk-donskaya-dasedokhin-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-dasedokhin-sw-4(config-if)#exit
msk-donskaya-dasedokhin-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-dasedokhin-sw-4(config)#line vty 0 4
msk-donskaya-dasedokhin-sw-4(config-line)#password cisco
msk-donskaya-dasedokhin-sw-4(config-line)#login
msk-donskaya-dasedokhin-sw-4(config-line)#exit
msk-donskaya-dasedokhin-sw-4(config)#line console
% Incomplete command.
msk-donskaya-dasedokhin-sw-4(config)#line console 0
msk-donskaya-dasedokhin-sw-4(config-line)#password cisco
msk-donskaya-dasedokhin-sw-4(config-line)#login
msk-donskaya-dasedokhin-sw-4(config-line)#exit
msk-donskaya-dasedokhin-sw-4(config)#enable secret cisco
msk-donskaya-dasedokhin-sw-4(config)#service password-encryption
msk-donskaya-dasedokhin-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-dasedokhin-sw-4(config)#ip domain-name donskeya-dasedokhin.rudn.edu
msk-donskaya-dasedokhin-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dasedokhin-sw-4.donskeya-dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dasedokhin-sw-4(config)#line vty 0 4
*Mar 2 22:30:25.817: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 2 22:30:25.817: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dasedokhin-sw-4(config-line)#transport input ssh
msk-donskaya-dasedokhin-sw-4(config-line)#exit
msk-donskaya-dasedokhin-sw-4(config)#exit
msk-donskaya-dasedokhin-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-dasedokhin-sw-4#
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left of the window, there is a "Top" button.

Рисунок 3.5: Настройка коммутатора msk-donskaya-dasedokhin-sw-4

3.6 слайд 6

```
msk-pavlovskaya-dasedokhin-sw-1
Physical Config CLI Attributes

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-pavlovskaya-dasedokhin-sw-1
msk-pavlovskaya-dasedokhin-sw-1(config)#interface vlan2
msk-pavlovskaya-dasedokhin-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-pavlovskaya-dasedokhin-sw-1(config-if)#no shutdown
msk-pavlovskaya-dasedokhin-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-dasedokhin-sw-1(config-if)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-dasedokhin-sw-1(config)#line vty 0 4
msk-pavlovskaya-dasedokhin-sw-1(config-line)#password cisco
msk-pavlovskaya-dasedokhin-sw-1(config-line)#login
msk-pavlovskaya-dasedokhin-sw-1(config-line)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#line console 0
msk-pavlovskaya-dasedokhin-sw-1(config-line)#password cisco
msk-pavlovskaya-dasedokhin-sw-1(config-line)#login
msk-pavlovskaya-dasedokhin-sw-1(config-line)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#enable secret cisco
msk-pavlovskaya-dasedokhin-sw-1(config)#service password-encryption
msk-pavlovskaya-dasedokhin-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-dasedokhin-sw-1(config)#ip domain-name pavlovskaya-dasedokhin.rudn.edu
msk-pavlovskaya-dasedokhin-sw-1(config)#crypto key generate rsa
% You already have RSA keys defined named msk-donskaya-dasedokhin-sw-1.donskaya-
dasedokhin.rudn.edu .
% Do you really want to replace them? [yes/no]: no
msk-pavlovskaya-dasedokhin-sw-1(config)#crypto key generate rsa
% You already have RSA keys defined named msk-donskaya-dasedokhin-sw-1.donskaya-
dasedokhin.rudn.edu .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: msk-pavlovskaya-dasedokhin-sw-1.pavlovskaya-
dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-dasedokhin-sw-1(config)#line vty 0 4
*Mar 1 0:33:33.186: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:33:33.186: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-dasedokhin-sw-1(config-line)#transport input ssh
msk-pavlovskaya-dasedokhin-sw-1(config-line)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#exit
msk-pavlovskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
w
Building configuration...
[OK]
msk-pavlovskaya-dasedokhin-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-dasedokhin-sw-1#
```

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Рисунок 3.6: Настройка коммутатора msk-pavlovskaya-dasedokhin-sw-1

4 Выводы

4.1 слайд 1

Я приобрел навыки по первоначальной настройке коммутаторов сети.